The optimal portfolio

A portfolio consists of a selection of different assets. The optimal portfolio achieves the ideal trade-off between potential risk and likely reward, depending on an investor's desired return and attitude to risk.

How it works

The optimal portfolio is a mathematical model that demonstrates that an investor will take on increased risk only if that risk is compensated by higher expected returns, and conversely, that an investor who wants higher expected returns must accept more risk.

The optimal portfolio reduces risk by selecting and balancing assets based on statistical techniques that quantify the amount of diversification between assets. A key feature of the optimal portfolio is that an asset's risk and return should not be assessed on its own, but by how it contributes to a portfolio's overall risk and return. The main objective of the optimal portfolio is to yield the highest return for a given risk or the lowest risk for a given return, these being investors' most common goals.

Efficient frontier

The efficient frontier is considered the optimum ratio between risk and reward – that is, the highest expected return for a defined level of risk, or the lowest risk for a given level of expected return. Portfolios that lie below the efficient frontier are sub-optimal, either because they do not provide enough return for that risk level, or because they have too high a level of risk for a defined rate of return. Asset correlation is a measure of the way investments move in relation to one another, and is important to the efficient frontier. A portfolio is better balanced if the prices of the securities in it move in different directions under similar circumstances, effectively balancing risk across the portfolio.